



SPACE DESIGN LOAD SUMMARY

2026-09-28

Studio 1_0

Level: Level 0 · Internal condition: Studio - Studio 1_0

DESIGN CRITERIA

	Heating	Cooling
Room set point	none	none
Outdoor dry bulb	—	—
Outdoor RH (coincident)	—	—
ROOM HUMIDITY		
Humidification set point	—	
lower RH limit		
Dehumidification set point	—	
upper RH limit		

SIZING (TAS DESIGN LOADS)

	Heating	Cooling
Design load	0 W	0 W
Design load per area	0.0 W/m ²	0.0 W/m ²
Sizing multiplier	not set	not set
Design loads: from Tas sizing — date/currency not recorded		

HEATING

Design day	Leeds_TRY ANN HTG 100% CONDS DB	
	Design day	Full year
Peak sensible load	2,268 W	802 W
Peak hour	15:00–16:00	1 Jan 00:00–01:00 (HOY 1)
Room dry bulb	15.1 °C	13.4 °C
Room resultant	13.2 °C	12.6 °C
Room RH	21 %	84 %
Room humidity ratio	2.2 g/kg	8.1 g/kg
Outdoor dry bulb	—	3.3 °C
Outdoor RH	—	100 %

SENSIBLE LOAD COMPONENTS AT PEAK

Component	Design day (W)	Full year (W)
Occupancy (sensible)	0	105
Equipment (sensible)	0	85
Infiltration / ventilation	-321	-154
Building heat transfer	-664	-327
External conduction — opaque	-895	-328
External conduction — glazing	-388	-184

Zero at both peaks (not listed): Solar, Lighting, Air movement (inter-zone), Air handling unit

LATENT GAINS AT THE SENSIBLE PEAK HOUR

Component	Design day (W)	Full year (W)
Occupancy (latent)	0	77

Zero at both peaks (not listed): Equipment (latent)

Latent gains: internal gains only (occupancy, equipment) at the hour of the sensible peak. The room's latent load, which also includes moisture from infiltration and ventilation, is not in the results and is not derived.

Components as the simulation reports them at each peak hour: + gain to the room air, – loss from it. No total is derived.

COOLING

Design day	Leeds_TRY ANN CLG 0% CONDS DB=>GRad	
	Design day	Full year
Peak sensible load	1,973 W	1,972 W
Peak hour	00:00–01:00	3 Jul 19:00–20:00 (HOY 4412)
Room dry bulb	19.8 °C	19.0 °C
Room resultant	21.5 °C	20.6 °C
Room RH	100 %	58 %
Room humidity ratio	14.5 g/kg	8.0 g/kg
Outdoor dry bulb	—	17.5 °C
Outdoor RH	—	59 %

SENSIBLE LOAD COMPONENTS AT PEAK

Component	Design day (W)	Full year (W)
Solar	0	836
Lighting	0	150
Occupancy (sensible)	105	150
Equipment (sensible)	85	450
Infiltration / ventilation	-9	-53
Building heat transfer	971	298
External conduction — opaque	898	219
External conduction — glazing	-77	-78

Zero at both peaks (not listed): Air movement (inter-zone), Air handling unit

LATENT GAINS AT THE SENSIBLE PEAK HOUR

Component	Design day (W)	Full year (W)
Occupancy (latent)	77	110

Zero at both peaks (not listed): Equipment (latent)

Latent gains: internal gains only (occupancy, equipment) at the hour of the sensible peak. The room's latent load, which also includes moisture from infiltration and ventilation, is not in the results and is not derived.

Components as the simulation reports them at each peak hour: + gain to the room air, – loss from it. No total is derived.

RESULTS

	Heating	Cooling
Status	Available	Available
Result source	SAM.Analytical.Tas	SAM.Analytical.Tas
Read into model	27 Sep 2026 20:42	27 Sep 2026 20:42
Matches current model	not recorded	not recorded

Peak hour: a design-day peak has an hour of the day only, no date; a full-year peak shows day and hour, no year, and its hour of the year (HOY 1 = 1 Jan 00:00–01:00, HOY 8760 = 31 Dec 23:00–24:00).

Design loads (sizing) and simulated peaks are shown side by side for information; no acceptance rule is applied.